Do Now · retrieval

Lesson 139 · SQL INSERT, UPDATE and DELETE Safely

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the difference between WHERE and ORDER BY in a query?

Q2

What are the outputs of a half adder when A=1 and B=1?

Q3

In Big O notation, what is binary search worst-case complexity?

Q4

When does a linear search stop?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Create rows — INSERT INTO Table(columns) VALUES(...).

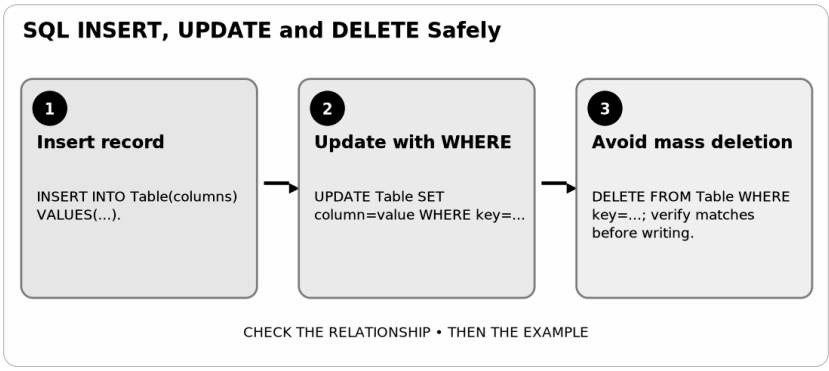
Core idea 2

Change rows safely — UPDATE Table SET column=value WHERE key=...

Core idea 3

Delete deliberately — DELETE FROM Table WHERE key=...; verify matches before writing.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

INSERT — Adds new rows.

UPDATE — Changes values in existing rows.

DELETE — Removes matching rows.

WHERE — Restricts affected rows; omission can affect every row.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A Book table has BookID=8,Stock=4. Write an UPDATE to set Stock=3 for that row.

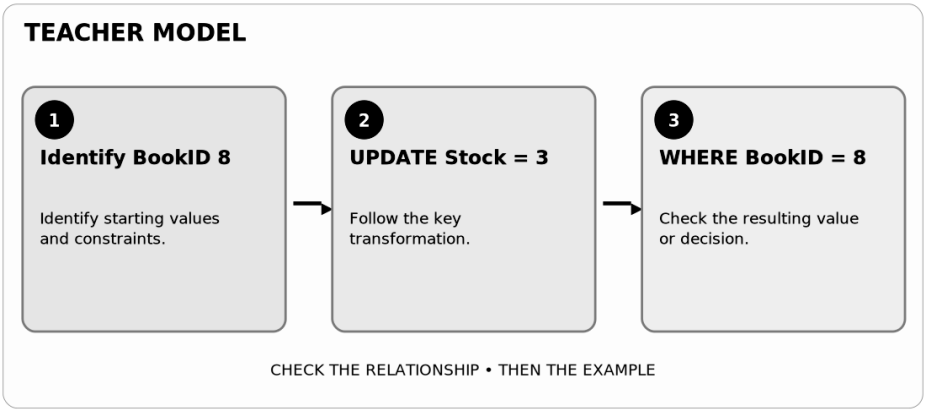
Follow the worked method

1. Identify BookID 8 — Find the given data and name the required result.

2. UPDATE Stock = 3 — Explain why this intermediate stage is necessary.

3. WHERE BookID = 8 — UPDATE Book SET Stock = 3 WHERE BookID = 8; Before running, SELECT the matching row. Without WHERE, the stock column of all books would be changed.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Insert club 24
Find the given data and name the required result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Write SQL inserting StudentID 22 and Name 'Kai' into Student. [4]

Question 2

A teacher must change a single pupil's YearGroup to 13. Explain a safe SQL statement. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain risks in DELETE FROM Booking; and suggest protections in a relational production database. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

UPDATE Book SET Stock=0; changes exactly one Book record.

Use key vocabulary from memory

1. Explain INSERT in your own words.

2. Explain UPDATE in your own words.

3. Explain DELETE in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.